

# Product Specification Of Low Loss Phase Stable Cable

# ST-206

## Structure Drawing



## Structure Characteristics

| Structure       | Item              | Standard Value                          |
|-----------------|-------------------|-----------------------------------------|
| Inner Conductor | Material          | Silver Plated Copper                    |
|                 | Construction (mm) | 1/1.45                                  |
|                 | Nom.Dia (mm)      | 1.45                                    |
| Insulation      | Material          | LD-PTFE                                 |
|                 | Nom.Dia (mm)      | 4.30                                    |
| Inner shield    | Material          | Flat Silver Plated Copper Wrapped Strip |
|                 | Nom.Dia (mm)      | 4.50                                    |
| Outer shield    | Material          | Round Silver Plated Copper Braided Wire |
|                 | Nom.Dia (mm)      | 4.95                                    |
| Jacket          | Material          | Aqua FEP                                |
|                 | Nom.Dia (mm)      | 5.40                                    |

## Electrical Characteristics

| Item                                      | Standard Value | Attenuation and Power handing@25℃&Sea Level |                       |                   |
|-------------------------------------------|----------------|---------------------------------------------|-----------------------|-------------------|
| Impedance (Ω)                             | 50             | Frequency (MHz)                             | Attenuation (dB/100m) | Power handing (W) |
| Capacitance (pF/m)                        | 87             | 500                                         | 15.46                 | 1509              |
|                                           |                | 1000                                        | 22.21                 | 1050              |
| Velocity of Propagation                   | 77%            | 2400                                        | 35.40                 | 659               |
|                                           |                | 3000                                        | 39.96                 | 584               |
| Max Operating Frequency (GHz)             | 25             | 6000                                        | 58.58                 | 398               |
|                                           |                | 8000                                        | 68.91                 | 339               |
| Screening Effectiveness (dB)              | >90            | 10000                                       | 78.29                 | 298               |
|                                           |                | 12400                                       | 88.67                 | 263               |
| Dielectric Withstanding Voltage (V@DC)    | 2500           | 18000                                       | 110.44                | 211               |
|                                           |                | 20000                                       | 117.62                | 198               |
| Phase Change VS Bending (°GHz)            | ±0.2           |                                             |                       |                   |
|                                           |                | K1                                          | 0.66491391400         |                   |
| Phase Change VS Temperature@ (-45~+85) °C | ≤1500 PPM      | K1                                          | 0.00117949500         |                   |
|                                           |                | IL=K1* √ FMhz+K2*FMHz                       |                       |                   |

## Mechanical &Environmental Characteristics

## Replaceable Type

|                              |              |            |         |
|------------------------------|--------------|------------|---------|
| Min Bending Radius (Static)  | 27.00 mm     | Micro-Coax | UFA210A |
| Min Bending Radius (Dynamic) | 54.00 mm     |            |         |
| Weight                       | 0.072 kg/m   |            |         |
| Operating Temperature        | -55~ +165 °C |            |         |